

DEEP LEARNING

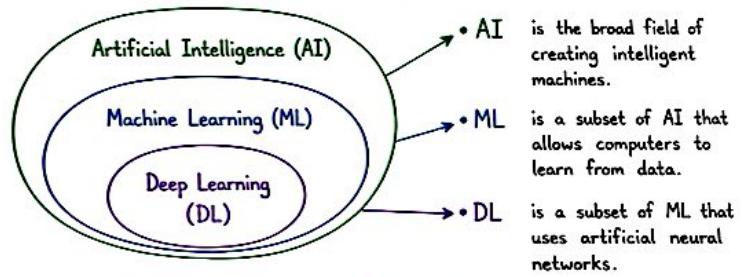
1. Definition

Deep Learning (DL) is a branch of Machine Learning that enables computers to learn by using Artificial Neural Networks, which are designed to work in a way similar to the human brain.

Deep Learning is especially useful for solving complex problems involving images, speech, text, and videos.



2. Relationship Between AI, ML, and DL



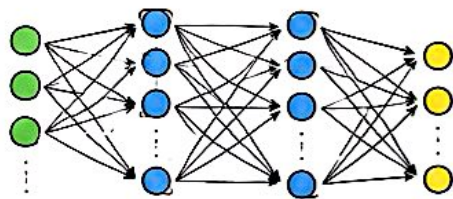
3. What is an Artificial Neural Network (ANN)?

An Artificial Neural Network (ANN) is a computer model inspired by the human brain.

It consists of many connected processing units called neurons, which work together to process information and learn patterns.

Structure of an ANN

Input Layer → Hidden Layer(s) → Output Layer



Input Layer

• Receives information.

Hidden Layer(s)

• Processes and learns patterns.

Output Layer

• Produces the final result.



Why is it called "Deep" Learning?

It is called Deep Learning because the neural network contains multiple hidden layers.

4. Characteristics of Deep Learning

- ✓ Learn very complex patterns
- ✓ Process large amounts of data
- ✓ Improve with more training data
- ✓ Automatically identify important features
- ✓ Handle images, speech, text, and videos

5. Applications of Deep Learning

	Image Recognition	• Face recognition • Medical image analysis • Object detection
	Speech Recognition	• Voice assistants • Voice typing • Speech-to-text systems
	Natural Language Processing (NLP)	• Language translation • Chatbots • Text summarization
	Self-Driving Vehicles	• Detect roads • Recognize traffic signs • Identify pedestrians
	Healthcare	• Disease detection • Medical image analysis • Drug research

6. Deep Learning vs Machine Learning

Feature	Machine Learning (ML)	Deep Learning (DL)
Relation	Subset of AI	Subset of Machine Learning
Model Used	Uses simpler models (e.g., Linear Regression, Decision Trees)	Uses Artificial Neural Networks (ANNs)
Feature Selection	Requires more manual feature selection	Learns features automatically
Data Requirement	Works well with smaller datasets	Performs best with very large datasets
Training Time	Faster to train	Slower to train
Best For	Simple to moderate problems	Complex problems such as images, speech, and videos

7. Advantages

- ✓ High accuracy for complex tasks
- ✓ Learns automatically from large datasets
- ✓ Excellent for image and speech recognition
- ✓ Can solve problems that are difficult for traditional programming
- ✓ Improves performance with more data



8. Limitations

- ✗ Requires large amounts of data
- ✗ Needs powerful computers (GPUs)
- ✗ Training can take a long time
- ✗ Difficult to understand how decisions are made ("black box")
- ✗ Requires significant computing resources



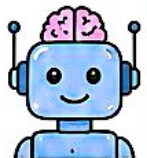
9. Key Terms

- Deep Learning (DL)**: A branch of Machine Learning that uses artificial neural networks with multiple hidden layers.
- Artificial Neural Network (ANN)**: A computer model inspired by the structure of the human brain.
- Neuron**: A small processing unit within a neural network that receives and passes information.
- Hidden Layer**: The layer where the neural network processes information and learns patterns.
- GPU (Graphics Processing Unit)**: A specialized processor that speeds up Deep Learning training.



10. Summary

- Deep Learning is a branch of Machine Learning.
- It uses Artificial Neural Networks inspired by the human brain.
- Deep Learning contains multiple hidden layers.
- It is widely used for image recognition, speech recognition, language processing, healthcare, and autonomous vehicles.
- Deep Learning performs very well on complex tasks but requires large amounts of data and computing power.

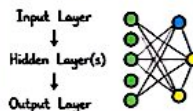


11. Quick Revision

Core Idea

- Deep Learning uses many layers to learn complex patterns from data.

ANN Structure



Common Applications

- ✓ Face Recognition
- ✓ Voice Assistants
- ✓ Language Translation
- ✓ Self-Driving Cars
- ✓ Medical Image Analysis

12. Memory Trick

AI (Big Box)

ML (Smaller Box)

DL (Smallest Box)

AI is the big field.
ML learns from data.
DL uses deep neural networks to learn very complex things!

